

Build and explore a polygenic index

30-minute lab, session 2

September 2026

Goal. Compute a PGI with PLINK from external-style weights, check its coverage, standardize it, examine its distribution, and read two group comparisons correctly. All supplied data are synthetic. Work in the terminal of your running Workbench JupyterLab app, in the same `aou_cli_workshop` folder as session 1.

0–5 minutes: orientation and preflight

```
cd aou_cli_workshop
bash scripts/00_preflight.sh
ls results/qc.pgen || bash scripts/02_qc.sh    # session-1 QC, if missing
head -n 3 data/pgi_weights.txt
head -n 3 data/groups.tsv
```

The weight file plays the role of published external discovery-GWAS weights (in real work: e.g. the PGI Repository). Count its columns — you will tell PLINK which is the variant ID (1), which is the *effect allele* (3), and which is the weight (5). `groups.tsv` carries two labels per person: `treatment` (randomized) and `enrolled` (self-selected).

5–12 minutes: score the PGI

```
less scripts/07_score_pgi.sh    # read it first; q exits
bash scripts/07_score_pgi.sh
head -n 3 results/pgi.sscore
grep -E "processed|skipped" results/pgi.log
```

Write your checkpoint: weight rows in the file ____; variants processed ____; skipped ____.

Name both reasons a row can be skipped: _____

12–20 minutes: explore

```
bash scripts/08_pgi_eda.sh
cat results/pgi_summary.txt
```

Open the three PNG figures by double-clicking them in the JupyterLab file browser (refresh the browser pane if they do not appear): `results/pgi_hist_treatment.png`, `pgi_kde_enrolled.png`, `pgi_decile_trait.png`.

Number	Meaning
$\text{corr}(\text{PGI}, \text{trait})$	Association between the index and the outcome
R^2 alone	Variance explained ignoring covariates — flattering
Incremental R^2	What the PGI adds <i>beyond</i> covariates — the honest number
Standardized difference	Group gap in SD units of the PGI

20–26 minutes: interpret with a partner

1. Why is incremental R^2 (0.063) so much smaller than R^2 alone (0.141)?
2. The randomized arms differ by -0.008 SD. Null result, or finding?
3. The enrolled groups differ by $+0.46$ SD. Say the honest causal sentence.
4. The decile figure is not monotonic everywhere. Problem, or expected?

5. If you drew this figure for a 300-person subgroup in ventiles, what rule starts to bite?

26–30 minutes: save and finish

```
bash scripts/04_save.sh
```

The same save as session 1: the disclosure screen runs first, and the archive now includes your PGI outputs and figures. Verify the cloud listing, then stop the app in the Workbench.

Reference answers: synthetic dataset only

2,001 weight rows; **1,922 processed, 79 skipped** — variants that session-1 QC removed from our files, plus 15 rows whose rsID-style names our data never contained (an ID-scheme mismatch, silent by design).

The PGI standardizes to mean 0, SD 1 by construction. $\text{corr}(\text{PGI}, \text{trait}) \approx 0.375$; R^2 alone ≈ 0.141 ; covariates alone ≈ 0.255 ; together ≈ 0.318 ; **incremental** $R^2 \approx 0.063$. The gap exists because the PGI shares signal with the PCs and other covariates — which is exactly why incremental R^2 is the reporting convention.

Treatment (randomized): -0.008 SD — a finding, not a failure: randomization balanced the PGI, and the figure doubles as a design check. **Enrolled** (self-selected): $+0.46$ SD — enrollment did not alter anyone's genome; self-selection correlates with population structure that leaked into the discovery weights. The correct sentence names selection and the weights, not biology. Non-monotonic deciles are expected sampling noise at ~ 59 people per decile; the SE bars say so. Ventiles of a 300-person subgroup average 15 people per cell — inside the 1–20 range the AoU dissemination policy restricts.

If you change things (encouraged, after the baseline run)

Each script ends with a TRY IT. Good ones here: change the KDE bandwidth in `scripts/pgi_eda.py` and decide which version you would publish; score the un-QC'd data (`07_score_pgi.sh`'s TRY IT) and explain the coverage change; re-draw the histogram by `sex`.

Quick command reference

<code>-score FILE 1 3 5</code>	Score: ID col 1, effect allele col 3, weight col 5
<code>header</code>	First weight-file line is names
<code>cols+=scoresums</code>	Also report raw score sums
<code>list-variants</code>	Write which variants entered the score
<code>.sscore</code>	One row per person: allele counts, score
<code>.sscore.vars</code>	The matched variant list (provenance)
<code>results/*.png</code>	Figures; open from the JupyterLab file browser

Sources and next steps

Weights for real traits: the PGI Repository (Becker et al. 2021, *Nat. Hum. Behav.*). Portability across ancestries: Martin et al. 2019 (*Nat. Genet.*). PLINK `-score`: cog-genomics.org/plink/2.0/score. Session-1 sources remain in `docs/SOURCES.md`.